

Geometric Competition Predicts Ambiguity in Language Model and Neural Emotion Representations

Anonymous authors - draft prepared for NeurIPS-style submission

Abstract

High-dimensional representations in both large language models and neural systems are often described geometrically, but the relationship between geometry and uncertainty remains underspecified. A common intuition is radial: samples farther from a class prototype should be more ambiguous. We test this assumption and show that it is incomplete. Across emotion-labelled language model representations, absolute distance from a class centroid captures only part of the structure, whereas relative competition with the nearest alternative centroid provides a direct account of ambiguity. We formalise this as a centroid margin and evaluate it against model logits, low-rank affective subspaces, and cross-validated uncertainty in human fMRI-derived region-of-interest representations. In fine-tuned language model embeddings, samples that fall closer to an incorrect centroid are assigned logits favouring that competing class in almost all overlap cases, and geometric margin is tightly coupled to logit margin. In the five-emotion brain representation, leave-one-subject-out decoding confirms above-chance emotion information, and margin predicts classifier uncertainty substantially better than raw distance; this relationship collapses under feature shuffling. We further introduce prototype purity as a class-level measure of representational exclusivity, showing that emotion categories differ in separability without requiring claims about psychological purity. Together, these results support a local geometric principle: ambiguity is not simply distance from a prototype, but relative competition among prototypes. The framework provides a reusable bridge between model interpretability and neural representational analysis while avoiding claims that artificial and biological systems share identical global organisation.

1 Introduction

Understanding how internal representations encode uncertainty is a central problem for machine learning, cognitive science, and computational neuroscience. Modern language models map text into high-dimensional spaces in which semantic categories appear as structured regions, while fMRI pattern analysis represents cognitive states as distributed activity across brain regions. These two settings differ profoundly in substrate and objective, yet both invite the same operational question: when a representation is ambiguous, what aspect of its geometry makes it ambiguous?

A tempting answer is radial. If each category has a prototype or centroid, then samples far from the centroid should be less reliable and samples near the centroid should be more certain. This account is intuitive, simple, and common in informal interpretations of embedding spaces. However, it ignores the fact that ambiguity is not defined by distance to one prototype alone. A sample can be far from its class centroid but still farther from every competitor, or close to its class centroid but equally close to another. Ambiguity therefore depends on relative competition between prototypes.

We study this issue in emotion representations because emotion categories are simultaneously discrete for modelling purposes and graded in representational structure. Classes such as happiness, love, fear, anger, sadness, and surprise are not expected to form perfectly isolated islands in natural language embeddings. Human neural responses to emotional stimuli are also distributed and overlapping rather than one-hot codes. This makes emotion a useful test bed for distinguishing simple cluster geometry from decision-relevant representational competition.

Our central claim is deliberately local and operational: under the tested conditions, ambiguity is better predicted by centroid competition than by absolute distance from a centroid. We do not claim that language models replicate the brain, that emotion categories are universal in their geometry, or that mathematical structure alone determines subjective affect. Instead, we propose a representation-level framework for quantifying ambiguity in any labelled high-dimensional space.

The paper makes four contributions. First, we define a centroid-margin account of sample-level ambiguity and contrast it against the radial distance hypothesis. Second, we show in language model embeddings that geometric overlap is tightly coupled to logits, linking embedding geometry to model decisions rather than to post-hoc visualisation alone. Third, we validate the same margin principle in fMRI-derived emotion representations using leave-one-subject-out decoding, bootstrap uncertainty, permutation controls, and feature shuffling. Fourth, we introduce prototype purity, a class-level metric that quantifies the proportion of samples occupying centroid-exclusive regions, allowing classes to be compared in separability without implying psychological or ontological purity.

Figure 1. Conceptual overview

Panel A: radial distance hypothesis. Panel B: centroid margin: true centroid distance minus nearest competing centroid distance. Panel C: class-level prototype purity. Panel D: two-system validation path: LLM logits and brain classifier uncertainty.

Leave blank in manuscript draft; replace with final rendered panel before submission.

Figure 1 should make the paper intelligible before the reader reaches the methods. The visual should explicitly distinguish radial distance, margin, overlap, and prototype purity. It should also show that purity is a class-level separability statistic, while margin is a sample-level ambiguity statistic.

2 Related Work**Representation geometry in language models**

Language model embeddings are commonly analysed through geometric structure: distances, directions, neighbourhoods, clusters, and low-dimensional subspaces. Work on probing, semantic directions, anisotropy, and mechanistic interpretability has shown that model representations encode task-relevant structure that can often be recovered by simple linear methods. However, geometric separability alone does not explain how ambiguous samples relate to model decisions. A cluster may be visually separated while its boundary regions still determine errors. Our work focuses on that boundary regime and asks whether local prototype competition predicts decision behaviour.

Neural representational geometry

In neuroscience, representational similarity analysis and multivariate pattern analysis treat patterns of neural activity as points in a representational space. Emotion fMRI data are especially suited to this approach because emotional states are distributed across cortical and subcortical networks, and class labels rarely imply clean separability. Prior work establishes that neural activity can contain decodable affective information, but the link between geometry and uncertainty is less direct. We therefore treat brain data as a biological validation setting for a geometric principle, not as evidence that artificial and neural systems are globally equivalent.

Margins, uncertainty, and prototype competition

Margins are central in classification theory, where decision confidence depends on relative separation between alternatives. Our contribution is to apply this idea to representation geometry itself. Rather than define uncertainty by the distance from a point to its own centroid, we define ambiguity by the distance to the true centroid relative to the nearest competing centroid. This converts ambiguity from a radial quantity into a relational quantity. It also creates a natural bridge between embedding geometry, classifier uncertainty, and logits.

3 Problem Formulation

Let x_i in $\mathbb{R}^d$ denote a representation vector for sample i , with label y_i in $\{1, \dots, K\}$. For language models, x_i is an embedding or hidden-state representation of an emotion-labelled text. For the brain dataset, x_i is a 48-dimensional vector of standardised region-of-interest features for one subject under one emotion condition.

Class centroids

For each class k , the centroid c_k is the mean representation of all samples labelled k .

$$c_k = (1 / N_k) * \sum_{i: y_i = k} x_i$$

Centroids are used as prototypes, not as literal pure emotional states. A centroid summarises the average representation of a class under a particular model, dataset, and feature construction.

Distance and the radial hypothesis

The radial hypothesis states that ambiguity should increase with own-centroid distance $d_{\text{true}}(i) = \|x_i - c_{\{y_i\}}\|_2$. Under this view, samples near the centroid are certain and samples far from the centroid are ambiguous. We test this directly rather than assuming it.

$$d_{\text{true}}(i) = \|x_i - c_{\{y_i\}}\|_2$$

Centroid margin

We define the nearest competing distance $d_{\text{comp}}(i)$ as the minimum distance between x_i and any centroid that is not its assigned class. The ambiguity margin A_i is own distance minus competitor distance.

$$d_{\text{comp}}(i) = \min_{k \neq y_i} \|x_i - c_k\|_2$$

$$A_i = d_{\text{true}}(i) - d_{\text{comp}}(i)$$

With this convention, larger A_i indicates stronger competition or misalignment: the sample is close to, or closer to, a competing centroid. Negative values indicate that the assigned centroid remains closer than the nearest competitor. Values near zero indicate boundary competition. The sign convention must be fixed throughout all figures and tables; if the paper uses the sign-reversed separation margin, every interpretation should be correspondingly reversed.

Prototype purity

Margin is a sample-level measure. We also define a class-level separability measure, prototype purity, to quantify how often samples from a class occupy a centroid-exclusive region. For class k , purity is the proportion of samples whose nearest centroid is their own class centroid.

$$P_k = (1 / N_k) * \sum_{i: y_i = k} 1[\|x_i - c_k\|_2 < \min_{j \neq k} \|x_i - c_j\|_2]$$

This is not a psychological claim that an emotion is pure. It is an operational statistic: a higher value means that, in the analysed representation, more samples of a class are closer to their own prototype than to any competing prototype. It can reveal that categories such as happiness or love are comparatively more separable than fear or anger, without implying that those emotions are intrinsically purer.

4 Experimental Design**4.1 Language model representations**

The language-model side evaluates emotion-labelled text embeddings across model families and training states. The intended core comparison includes BGE and MPNet variants, with base and fine-tuned checkpoints, and hidden representations from middle and final layers. Each sample is assigned one of six emotion labels: anger, fear, happiness, love, sadness, or surprise. The reported validation set contains 4038 samples. Embeddings are extracted using a fixed pooling procedure and evaluated in the same metric space for each

model variant.

The LLM pipeline has five phases. Phase 1 quantifies class topology using silhouette and centroid separation. Phase 2 measures overlap and radial structure. Phase 3 removes discriminative directions to test whether emotional information is low-rank. Phase 4 projects into a compact affective subspace and tests whether classification signal is retained. Phase 5 links geometric overlap directly to logits. This ordering is important: clustering is not treated as the main contribution; the decisive evidence is whether geometric competition predicts model decisions.

Each phase should be implemented with the same preprocessing choices across models to avoid confounding geometry with pipeline differences. The final manuscript should specify whether embeddings were L2-normalised before Euclidean distances were computed, how text was tokenised, whether hidden states were mean-pooled or CLS-pooled, and whether logits were read from the fine-tuned classifier head or reconstructed from a probe. These details are not cosmetic; a reviewer can reasonably reject a geometry paper if distance metrics, pooling, or normalisation differ across conditions without justification.

The LLM experiment should also report the unit of resampling for uncertainty estimates. For statistics over text samples, bootstrap resampling should occur over validation examples. For class-level metrics such as prototype purity, resampling should preserve class labels or resample within class so that uncertainty reflects sample variability rather than class-frequency differences.

Figure 2. LLM geometry across training state and layer

Insert silhouette ranking, overlap rates, and illustrative 2D projection. Must show base/mid-layer spaces as diffuse and fine-tuned final layers as sharply separated. Include exact values in panel labels.

Leave blank in manuscript draft; replace with final rendered panel before submission.

4.2 Geometry-logit coupling

For each LLM sample, we compute geometric margin and compare it with logit margin. Overlap cases are samples whose nearest centroid belongs to a class other than their true label. If geometry is merely a visual artefact, overlap should not systematically correspond to logits. If geometry is decision-relevant, overlap cases should tend to be assigned higher logits for the competing class.

The key statistics are: overlap count, overlap percent, logit agreement rate, average logit difference in overlap cases, and distance-logit correlation. p-values from permutation or bootstrap procedures should be reported as $p < 0.001$ or as a permutation lower bound, never as $p = 0.0$.

4.3 Low-rank affective subspace

To test whether the observed emotion geometry is distributed throughout the full embedding space or concentrated in a compact set of directions, we use linear classifiers and

singular-vector-based direction removal. If removing a small number of directions collapses classification performance, the affective signal is concentrated. Conversely, if performance declines slowly, the signal is redundantly distributed. A 20-dimensional affective subspace is then tested for accuracy retention and silhouette improvement.

4.4 Brain representation

The neural validation uses the NeuroEmo five-emotion fMRI dataset from OpenNeuro accession ds005700. The analysed representation contains 40 healthy participants and five emotion conditions: Afraid, Calm, Delighted, Depressed, and Excited. Each subject contributes one vector per emotion after averaging timepoints aligned to the relevant emotional blocks. The final matrix contains 200 subject-emotion observations with 48 Harvard-Oxford cortical ROI features plus subject and emotion identifiers.

The brain dataset is used for biological validation of the geometry principle, not for claiming full equivalence between neural and language-model representations. The brain labels differ from the LLM labels, and the fMRI features are lower-dimensional, noisier, and subject-averaged. For this reason, the strongest cross-system claim is local: margin-based competition predicts uncertainty in both systems.

The ROI representation should be described as a derived feature table rather than as raw fMRI. This matters because the statistical object analysed by the paper is not a voxel image, but a subject-by-emotion matrix. Each row corresponds to one participant under one emotion condition, and each feature corresponds to a cortical region extracted with a labelled atlas. This construction gives interpretability and tractability, but it also compresses trial-level variability. The limitation should be acknowledged explicitly rather than hidden.

Figure 3. Brain processing and validation pipeline

Insert BIDS/task-fe to ROI extraction to subject x emotion matrix. Show 40 subjects x 5 emotions = 200 samples, 48 ROI features, LOSO validation, margin analysis, and feature shuffle control.

Leave blank in manuscript draft; replace with final rendered panel before submission.

4.5 Brain uncertainty validation

Because the brain dataset does not contain model logits or subjective confidence scores for each subject-emotion point, uncertainty is estimated through cross-validated classifier probabilities. We use leave-one-subject-out validation: all five observations from one subject are held out together, and the classifier is trained on the remaining subjects. This prevents leakage across subject identity and tests whether emotion structure generalises to unseen individuals.

For each held-out sample, classifier uncertainty is defined as 1 minus the predicted probability assigned to the true class. We then compare uncertainty against own-centroid distance and centroid margin. A feature-shuffle control destroys multivariate structure while preserving marginal feature distributions; if the margin-uncertainty relationship disappears, the effect depends

on real representational structure rather than artefactual class labels or univariate scale differences.

5 Results

5.1 LLM emotion geometry sharpens with supervision

Fine-tuned final-layer embeddings form much clearer affective geometry than base or middle-layer representations. Reported silhouette scores show MPNet-FT-Final as the strongest separated space (0.8597), followed by BGE-FT-Final (0.6847), while base and middle-layer spaces remain close to zero. This establishes that supervised affective tuning sharpens emotion clusters, but this result alone is not the central contribution. It primarily sets up the more important question: whether the resulting geometry is coupled to decisions and uncertainty.

Overlap rates follow the same pattern. Fine-tuned final-layer models show low overlap, while base and middle-layer spaces contain more boundary entanglement. Reported global overlap includes BGE-Base-Final at 19.22%, MPNet-Base-Final at 18.45%, BGE-FT-Final at 1.81%, and MPNet-FT-Final at 0.42%. This demonstrates that fine-tuning reduces overlap, but also leaves rare boundary cases that are highly informative for testing whether geometry predicts logits.

Figure 4. Training state and overlap structure

Panel A: silhouette ranking across all model variants. Panel B: overlap percentage by model and layer. Panel C: radial zones showing core, transition, and high-overlap boundary. Panel D: pairwise confusion heatmap, especially Happiness-Love and Fear-Anger.

Leave blank in manuscript draft; replace with final rendered panel before submission.

Pairwise confusion structure should be used to make this section more than a global clustering result. The reported pretrained spaces show stronger entanglement between semantically related affective categories, particularly happiness-love and fear-anger, than between more distant classes. This is important because it shows that overlap is not random noise. Boundary structure follows interpretable affective relationships, which is exactly what a prototype-competition account predicts.

5.2 Distance alone is not a sufficient ambiguity account

The naive radial hypothesis predicts that own-centroid distance should explain uncertainty. In the brain validation, this relationship is weak: own-centroid distance has Spearman $r = 0.1509$ with classifier uncertainty, Pearson $r = 0.1471$, and AUC = 0.5627 for uncertainty discrimination. These values are above zero but insufficient to support a radial account. Distance captures some dispersion, but it does not tell us whether a point is near a competing class.

This negative result is strategically important. It prevents the paper from assuming that distance equals ambiguity and motivates the margin formulation. In the final manuscript, this should be presented as a falsified baseline rather than as a

failed experiment. The reader should see that the authors tested the simplest possible explanation and found it incomplete.

5.3 Centroid margin predicts ambiguity

Centroid margin provides a much stronger account of ambiguity because it measures competition. In the brain representation, margin correlates strongly with classifier uncertainty: Spearman $r = 0.6108$, Pearson $r = 0.6351$, and AUC = 0.8124. This means that samples positioned close to a competing centroid are substantially more likely to be uncertain than samples that are simply far from their own centroid.

The feature-shuffle control is essential. After shuffling features, the margin-uncertainty relationship collapses to $r = 0.0287$ with $p = 0.687$. This shows that the margin effect depends on the multivariate ROI structure and is not a trivial consequence of label frequencies, distance scaling, or the mere existence of centroids.

Figure 5. Margin predicts uncertainty while distance fails

Panel A: distance vs uncertainty scatter with weak trend. Panel B: margin vs uncertainty scatter with fitted trend and r /AUC values. Panel C: AUC comparison for distance vs margin. Panel D: feature-shuffle collapse. Use identical axes where possible.

Leave blank in manuscript draft; replace with final rendered panel before submission.

5.4 Geometry is coupled to logits in language models

The strongest LLM evidence comes from directly comparing geometric competition with logits. In fine-tuned final-layer models, overlap cases are rare but highly diagnostic. For BGE-FT-Final, 76 of 4038 validation samples fall closer to an incorrect centroid, giving an overlap rate of 1.882%. For MPNet-FT-Final, 71 of 4038 samples fall into overlap, giving 1.758%. In these cases, logits agree with the geometric competitor in 93.42% of BGE cases and 100% of MPNet cases.

The magnitude of logit difference also supports decision relevance. In overlap regions, the average logit difference favours the competing class by approximately -3.3245 for BGE and -3.4174 for MPNet under the reported sign convention. Most importantly, geometric margin and logit margin are extremely correlated: $r = 0.9572$ for BGE-FT-Final and $r = 0.9884$ for MPNet-FT-Final. These values indicate that embedding-space competition is not merely a visual or clustering artefact; it is tightly coupled to model decision behaviour.

Figure 6. Geometry-logit coupling

Panel A: definition of overlap and logit agreement. Panel B: overlap count/percent for BGE and MPNet. Panel C: agreement rate bars. Panel D: geometric margin vs logit margin scatter with $r = 0.957$ and $r = 0.988$. Panel E: selected overlap examples if available.

Leave blank in manuscript draft; replace with final rendered panel before submission.

5.5 Emotion information is concentrated in a compact subspace

Direction-removal experiments show that emotion-class information is not uniformly spread through the full embedding space. Baseline linear decoding is high in final-layer embeddings, with reported accuracies around 98% for fine-tuned BGE and MPNet. Removing the top discriminative directions causes a rapid collapse, especially in fine-tuned models. After removing 10 dimensions, MPNet-FT drops to 51.98% and BGE-FT to 63.24%; after 15 to 20 removed directions, both approach chance.

This supports a low-rank affective core. When projected into a 20-dimensional affective subspace, accuracy is retained or slightly improved, while silhouette improves substantially. Reported silhouette changes include MPNet-FT from 0.8597 to 0.9068, BGE-FT from 0.6847 to 0.8353, MPNet-Base from 0.0471 to 0.4075, and BGE-Base from 0.0570 to 0.4146. The subspace therefore denoises the representation and isolates the directions most relevant to affective class structure.

Figure 7. Low-rank affective structure

Panel A: accuracy decay after removing top discriminative directions. Panel B: erasure point by model. Panel C: 20D silhouette improvement. Panel D: accuracy retained in 20D. Keep full curves in appendix and a compressed main-panel summary here.

Leave blank in manuscript draft; replace with final rendered panel before submission.

5.6 Brain emotion representations contain decodable structure

Before interpreting brain geometry, we verify that the fMRI-derived ROI space contains emotion-related information. Leave-one-subject-out decoding yields accuracy = 0.5600 and balanced accuracy = 0.5600, compared with a five-class chance level of 0.2000. Bootstrap 95% confidence intervals are [0.4900, 0.6300] for accuracy and [0.4906, 0.6280] for balanced accuracy. A permutation null has mean 0.1978 and SD 0.0334, with $p < 0.001$ under a standard permutation-bound reporting convention.

This result has two implications. First, the brain representation contains real emotion-related structure that generalises to

held-out subjects. Second, the moderate accuracy is not a weakness for the geometry argument. Unlike fine-tuned language models, fMRI ROI representations are noisy and overlapping. This makes them useful for testing graded ambiguity rather than merely demonstrating perfect separability.

5.7 Prototype purity reveals class-level separability

Margin describes sample-level ambiguity, but classes also differ in how exclusively their samples occupy the region around their own centroid. Prototype purity quantifies this class-level separability. A high purity score means that many samples from a class are closer to their own centroid than to any competing centroid; a low score means that the class is frequently geometrically entangled with alternatives.

This metric allows the paper to discuss findings such as happiness and love being more separable without using unsupported psychological language. The correct interpretation is not that happiness or love are intrinsically purer emotions. Rather, under the analysed embedding geometry, these categories may occupy comparatively more centroid-exclusive regions, while other categories show stronger prototype competition. In the final results, report purity values by class with confidence intervals or bootstrap bands, and connect them explicitly to margin distributions and overlap rates.

Figure 8. Prototype purity and class-level separability

Panel A: prototype purity P_k by emotion class. Panel B: margin distributions by class. Panel C: overlap rate by pair. Panel D: relationship between class purity and mean logit margin. Ensure all claims say "in this embedding space".

Leave blank in manuscript draft; replace with final rendered panel before submission.

The purity analysis should include two safeguards. First, report purity together with sample counts and confidence intervals so that high purity is not confused with class imbalance. Second, report whether the ordering of class purity is stable across model variants. If happiness and love are consistently more centroid-exclusive in fine-tuned models, but not in base models, the interpretation becomes training-dependent rather than universal. If the ordering is stable across both, it supports a stronger claim about the semantic geometry of these labels in the tested representation families.

5.8 Local convergence despite global divergence

The paper should distinguish local geometric principles from global representational alignment. The brain centroid geometry appears strongly arousal-aligned in the first principal component: PC1-arousal $r = 0.965$, $p = 0.0078$, while PC1-valence is weak. This should be interpreted cautiously because the analysis uses five emotion centroids, but it provides a plausible explanation for why neural and LLM spaces may differ globally even if they share a local margin-ambiguity relation.

This distinction protects the central claim. We are not arguing that brain and LLM emotion maps are globally identical. Instead, we show that when representations are treated as labelled geometric spaces, ambiguity is better described by relative

competition than by absolute distance. Global axes can differ while local boundary logic remains comparable.

Figure 9. Global structure comparison

Panel A: brain centroid PCA with valence/arousal annotation. Panel B: brain RDM heatmap. Panel C: LLM RDM or affective subspace structure. Panel D: schematic showing global divergence plus local margin convergence. Treat as supporting, not central proof.

Leave blank in manuscript draft; replace with final rendered panel before submission.

6 Discussion

The central finding is that ambiguity in emotion representations is relational. A sample is not ambiguous merely because it is far from its own centroid; it is ambiguous when a competing centroid becomes geometrically plausible. This is why margin outperforms raw distance in the brain data and why geometric overlap aligns with logits in language models. The mechanism is therefore not clustering alone, but competition among class prototypes.

This distinction should drive the entire manuscript structure. If the paper is read as a clustering paper, the contribution will seem incremental because fine-tuned classifiers obviously separate labelled emotions. If it is read as a margin paper, the contribution becomes stronger: the authors show that the geometry of boundary competition predicts downstream decision behaviour and cross-validated uncertainty. The manuscript should therefore keep returning to the same question: does this result support the claim that ambiguity is governed by relative prototype competition?

This framing also resolves the issue of representational purity. It is tempting to say that no emotion exists purely as itself, but that would overstate what the data prove. The safer and more precise statement is that representations are not perfectly isolated: samples exhibit graded proximity to multiple prototypes, and classes vary in centroid exclusivity. Prototype purity provides a mathematically defined way to quantify this without making claims about subjective emotional ontology.

The cross-system aspect should be interpreted carefully. The LLM and brain representations differ in dimensionality, label set, measurement process, and noise. The strength of the paper is not a claim of one-to-one alignment between artificial and biological emotion spaces. Instead, it is the demonstration that the same style of local geometric measurement can connect ambiguity to decisions or uncertainty in both systems. This makes the framework useful for interpretability, not just for affective computing.

The low-rank results further suggest that emotion-relevant information can be concentrated into a compact subspace. In fine-tuned models, the affective signal becomes highly separable and vulnerable to removal of a small number of discriminative directions. In base models, the signal is weaker but can become clearer after subspace isolation. This raises an important future direction: whether model training changes not only the amount of emotion information but the geometry through which that information becomes decision-relevant.

Finally, the brain validation is deliberately modest but important. Above-chance LOSO decoding establishes that the ROI representation contains emotion structure. Margin-uncertainty coupling, and its collapse after feature shuffling, show that local prototype competition captures something meaningful about neural representations. This is not enough to claim a universal law of emotion, but it is enough to support a reusable geometric hypothesis that can be tested in additional datasets and modalities.

7 Limitations

First, the brain validation uses one primary five-emotion fMRI dataset. Although the subject-level design and LOSO validation are meaningful, broader neural datasets are needed to test generalisation across cultures, stimulus types, scanners, task designs, and emotion taxonomies.

Second, the LLM analysis depends on the chosen emotion-labelled text dataset, pooling procedure, model families, and layers. The results should be reported as applying to the tested BGE and MPNet variants unless additional architectures are evaluated.

Third, centroid methods assume that class means are useful prototypes. This assumption is reasonable for the present framework but not guaranteed for all representation spaces. Nonlinear manifolds, multimodal class distributions, or hierarchical emotion structures may require richer prototype definitions.

Fourth, the geometry-logit results show strong coupling, not causal proof. Without interventions that move samples in embedding space and measure resulting logit changes, causal language should be avoided. Direction removal provides partial intervention-like evidence, but the paper should use terms such as predicts, aligns with, or is coupled to rather than causes.

Fifth, valence-arousal analyses based on five centroids are underpowered and should remain supporting evidence. They are useful for explaining global differences, not for proving a broad theory of affective axes.

8 Reproducibility, ethics, and submission readiness

8.1 Reproducibility package

A NeurIPS submission built around representation geometry must be unusually transparent because small implementation choices can change distances and margins. The reproducibility package should therefore include scripts that regenerate every reported table from raw or public intermediate files. The minimum package should contain separate scripts for embedding extraction, LLM centroid geometry, low-rank direction removal, affective subspace construction, logit-consistency analysis, brain ROI geometry, brain cross-validation, and figure generation. The paper should not depend on manual spreadsheet edits or notebook state.

The released results should be saved in both human-readable and machine-readable forms. For example, the logit-consistency analysis should export a JSON file containing overlap count, overlap percentage, agreement rate, average logit difference, correlation coefficient, confidence interval, and p-value for each model. The brain analysis should export LOSO predictions, predicted probabilities, true labels, margins, own distances, competing distances, uncertainty, and shuffle-control summaries. This makes it possible for reviewers to verify the

reported statistics without rerunning the entire pipeline.

8.2 Ethical scope

The project analyses public, de-identified fMRI data and pretrained or fine-tuned language model representations. It does not collect new human-subject data, does not infer clinical status, and does not propose deployment for diagnosis or mental-health decision-making. The main ethical risk is interpretive: readers may overstate the similarity between LLMs and human brains or treat embedding geometry as evidence about subjective emotional experience. The manuscript should explicitly state that the framework concerns representation geometry, not the ontology of emotion or clinical affective assessment.

The paper should also avoid anthropomorphic language about language models. Terms such as uncertainty, ambiguity, and emotion should be defined operationally: uncertainty means logit or classifier uncertainty; ambiguity means geometric competition; emotion means dataset label. This precision protects the work from being read as claiming that models experience emotion.

8.3 Why this belongs at NeurIPS

The submission should be positioned as an interpretability and representation-learning paper rather than as a pure neuroscience report. Its value to the NeurIPS community is threefold. First, it provides a reusable method for linking embedding geometry to decision behaviour. Second, it shows how low-rank affective subspaces can be used to separate meaningful signal from high-dimensional noise. Third, it demonstrates how the same metric family can be evaluated in a biological representation space without requiring global brain-model equivalence.

The strongest reviewer-facing message is that the paper turns a descriptive visual intuition into a testable mechanism. Clusters and projections are common; the novel claim is that boundary competition predicts downstream uncertainty. This is why the logit and brain validation results must be central, while silhouette and projection plots should be treated as setup rather than proof.

8.4 Final claim hierarchy

The main text should maintain a clear hierarchy of claim strength. The strongest claim is that centroid margin is coupled to LLM logits in the tested fine-tuned models. The second strongest claim is that centroid margin predicts cross-validated uncertainty in the NeuroEmo ROI representation better than raw distance. The third claim is that fine-tuning compresses affective information into a compact subspace. The fourth, more cautious claim is that global brain and LLM organisation may differ even when local margin structure is comparable. Prototype purity should remain a supporting class-level analysis unless its values are robust across models and resampling procedures.

8 Conclusion

We introduced a geometric framework for ambiguity in emotion representations. Across language model embeddings and fMRI-derived neural representations, ambiguity is better captured by relative centroid competition than by absolute distance from a prototype. In LLMs, geometric overlap is tightly coupled to logits. In brain data, margin predicts classifier uncertainty under leave-one-subject-out validation and collapses after feature shuffling. Prototype purity complements this sample-level account by quantifying class-level separability.

These results support a cautious but powerful conclusion: emotion representations are organised by local geometric competition, and this principle can be studied across artificial and biological systems without claiming global equivalence.

References

- Kriegeskorte, N., Mur, M., and Bandettini, P. (2008). Representational similarity analysis - connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*.
- Haxby, J. V., Connolly, A. C., and Guntupalli, J. S. (2014). Decoding neural representational spaces using multivariate pattern analysis. *Annual Review of Neuroscience*.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G., and Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*.
- Ethayarajh, K. (2019). How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. *EMNLP-IJCNLP*.
- Mishra, S., Srinivasan, N., Asif, M., and Tiwary, U. S. (2023). Affective film dataset from India (AFDI): Creation and validation with an Indian sample. *Journal of Cultural Cognitive Science*.
- OpenNeuro. NeuroEmo: An fMRI Dataset for Emotion Recognition. Accession ds005700, version 1.2.0.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*.
- Abraham, A. et al. (2014). Machine learning for neuroimaging with scikit-learn. *Frontiers in Neuroinformatics*.
- Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*.
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

Appendix A. Full metric definitions

This appendix gives the precise operational definitions that should be kept consistent across code, figures, and text. The final paper should include the exact sign convention used in all tables.

```
c_k = (1 / N_k) * sum_{i: y_i = k} x_i
d_true(i) = ||x_i - c_{y_i}||_2
d_comp(i) = min_{k != y_i} ||x_i - c_k||_2
A_i = d_true(i) - d_comp(i)
u_i = 1 - p_hat(y_i | x_i)
P_k = (1 / N_k) * sum_i 1[nearest_centroid(x_i) = k], for i with y_i = k
```

For LLM logits, define z_{true} as the logit for the labelled class and z_{comp} as the logit for the nearest competing centroid class. The logit margin can be defined as $z_{\text{true}} - z_{\text{comp}}$. If the geometric margin is sign-reversed in the code, adjust the reported correlation sign accordingly.

Appendix B. Statistical reporting plan

All main claims should include uncertainty estimates or controls. For LLM geometry-logit correlations, report Pearson and Spearman correlations, confidence intervals from bootstrap resampling over samples, and permutation p-values from label or margin shuffles. For brain validation, report LOSO accuracy, balanced accuracy, bootstrap confidence intervals over subjects or subject-emotion samples, permutation null distributions, and feature-shuffle controls.

Claim	Statistic	Required control
Distance is weak	r, AUC	Compare to margin on same samples
Margin predicts uncertainty	r, AUC, CI	Feature shuffle and label permutation
Geometry predicts logits	r, agreement rate	Shuffle centroid labels or logits
Prototype purity differs by class	P_k with CI	Bootstrap samples within class
Low-rank affective core	Accuracy decay, erasure plot	Random direction removal baseline

Appendix C. Main figure checklist

- Figure 1 must introduce distance, margin, overlap, and prototype purity in one schematic.
- Figure 2 should establish that fine-tuning sharpens LLM emotion geometry but should not be treated as the main result.
- Figure 5 should be the critical brain validation figure: distance weak, margin strong, shuffle collapse.
- Figure 6 should be the critical LLM mechanism figure: overlap cases align with logits and geometric margin correlates with logit margin.
- Figure 8 should present prototype purity as a class-level separability metric, with confidence intervals if possible.
- All placeholder figure captions must be replaced by actual captions with exact statistics before submission.

Appendix D. Reproducibility checklist

- Release anonymised scripts for LLM embedding extraction, centroid geometry, logit consistency, subspace projection, brain ROI geometry, and statistical tests.
- Pin package versions for numpy, scipy, scikit-learn, PyTorch or Transformers, Nilearn, pandas, matplotlib, and any model-loading libraries.

- Document all random seeds, model checkpoint names, pooling strategies, hidden layers, normalisation, train/validation split definitions, and number of bootstrap/permutation iterations.
- Provide a data README describing public sources, derived matrices, licenses, and expected output files.
- Remove all author names, local file paths, machine usernames, and institution-identifying metadata for anonymous review.

Appendix E. Reviewer-risk audit

Risk	Reviewer concern	Protection in paper
Overclaiming brain-LLM equivalence	Systems differ too much	State local geometric comparability, not global equivalence
Purity language	Sounds philosophical	Define prototype purity mathematically and restrict to class
Only five brain emotions	Underpowered global geometry	Use brain as validation; keep VA as supporting
Correlation not causation	No intervention	Use coupled/predicts/aligns; avoid causes
Metric choice	Why Euclidean?	Add cosine/normalised sensitivity in appendix

Appendix F. Suggested table shells for final manuscript

The final version should include compact numerical tables in the appendix. The following tables are intentionally specified as shells so that exact final outputs can be inserted after code verification.

Table	Rows to include	Purpose
LLM phase summary	model, layer, training state, silhouette, overlap rate	Establish geometry across variants
Logit consistency	overlap count, agreement rate, logit difference	Centrist LLM anchor
Brain validation	LOSO accuracy, balanced accuracy, CI, null model	Validate neural signal
Margin comparison	distance r/AUC, margin r/AUC, shuffle r/p	Shows margin > distance
Prototype purity	emotion, purity, CI, overlap rate, mean margin	Class-level separability

Appendix G. Language for claim discipline

- Avoid:** LLMs replicate the brain
Use: LLMs and fMRI-derived representations show a comparable local margin-ambiguity relationship under the tested conditions
- Avoid:** No emotion exists purely as itself
Use: No class forms a perfectly isolated region under the defined prototype-purity metric
- Avoid:** Geometry causes logits
Use: Geometric margin is tightly coupled to, and predictive of, logit margin
- Avoid:** Universal law of emotion
Use: A reusable geometric principle observed across the tested models and dataset
- Avoid:** Happiness is purer than fear
Use: Happiness has higher prototype purity than fear in the analysed embedding space

Appendix H. NeurIPS checklist notes

- Claims: every abstract claim must map to a numbered result or figure.
- Limitations: include dataset scope, emotion-label scope, centroid assumption, no causal intervention, and global-axis caveat.
- Reproducibility: list exact model checkpoints, dataset version, scripts, seeds, and package versions.
- Statistical significance: central claims require confidence intervals, permutation tests, or bootstrap intervals.
- Ethics: public de-identified OpenNeuro data, no new human subjects, no deployment claims.

- Assets: if processed matrices or embeddings are released, include provenance and license notes.

Appendix I. Expanded figure captions to paste into the final paper

Figure 1. Centroid competition framework. A radial account treats ambiguity as a function of distance from the assigned centroid. The margin account instead compares the assigned centroid with the nearest alternative centroid. Prototype purity extends the same geometry from sample-level ambiguity to class-level separability.

Figure 2. Training and layer effects in LLM emotion geometry. Fine-tuned final-layer models show substantially higher silhouette and lower overlap than base and middle-layer representations, indicating that supervision sharpens the affective topology.

Figure 3. NeuroEmo brain pipeline. Raw BIDS fMRI task data are transformed into a 40 subject by 5 emotion ROI matrix using Harvard-Oxford cortical parcels. The derived 48D representation is used for decoding, centroid geometry, and uncertainty validation.

Figure 4. Overlap structure across LLM variants. Boundary entanglement is most visible in pretrained or middle-layer spaces and is reduced by fine-tuning. Pairwise overlap should be reported to show that confusion follows interpretable affective relationships rather than random collapse.

Figure 5. Brain margin validation. Own-centroid distance weakly predicts classifier uncertainty, while centroid margin strongly predicts uncertainty and yields higher AUC. Feature shuffling removes the margin relationship, indicating dependence on multivariate neural structure.

Figure 6. Embedding geometry is coupled to logits. Overlap samples are rare in fine-tuned final-layer spaces but highly informative: when a sample is closer to a competing centroid, logits usually favour that competitor, and continuous geometric margin tracks logit margin.

Figure 7. Affective information is low-rank. Removing top discriminative directions collapses emotion decoding, while a compact 20D affective subspace retains classification signal and improves geometric separability.

Figure 8. Prototype purity by emotion class. Purity is defined operationally as nearest-centroid exclusivity. Higher values indicate greater separability in the analysed representation, not intrinsic psychological purity.

Figure 9. Local convergence with global divergence. Brain and LLM spaces may differ in global valence-arousal organisation while sharing a local margin-ambiguity relation.

Appendix J. Pseudocode for reproducible analysis

1. Load emotion-labelled validation samples and extract embeddings for each model, layer, and training state using a fixed pooling strategy.
2. For each representation matrix, standardise or normalise features according to the pre-registered metric choice, then compute class centroids.
3. Compute own-centroid distance, nearest competing centroid distance, centroid margin, overlap indicator, and prototype purity for every class.
4. For LLMs with logits, compute true-class logit, nearest-competitor logit, logit margin, overlap agreement, and geometry-logit correlation.
5. Train linear probes to estimate decodability, then remove top discriminative directions iteratively to test concentration of affective information.
6. Construct the 20D affective subspace from discriminative directions and rerun silhouette, accuracy, overlap, and purity analyses.
7. For NeuroEmo, construct the subject-emotion ROI matrix, scale features, compute centroids, and run leave-one-subject-out decoding.

8. Estimate brain uncertainty as one minus cross-validated true-class probability and compare uncertainty against distance and margin.
9. Repeat the brain margin analysis after feature shuffling and repeat LLM margin-logit analysis after label or centroid permutation.
10. Export all statistics as machine-readable JSON/CSV and generate all manuscript figures from those exported files only.

Appendix K. Minimum acceptance-critical additions still required

- Insert the final prototype purity values by class. The draft currently defines the metric and interpretation, but the result must include exact values and uncertainty intervals.
- Add LLM bootstrap confidence intervals for geometry-logit correlations. Large r values are persuasive, but NeurIPS reviewers will expect uncertainty estimates or permutation tests.
- Add cosine-distance or L2-normalised sensitivity analysis, even if it is relegated to appendix. This pre-empts the common criticism that Euclidean geometry is arbitrary.
- Verify the sign convention for every margin value. A sign error in one figure caption can make the whole mechanism look confused.
- Confirm that the cross-system comparison does not rely on an unverified three-emotion RSA result. If that result cannot be traced to source files, exclude it from the main paper.
- Replace every placeholder figure with a vector-quality PDF or high-resolution raster. The current boxes indicate where figures belong but are not submission figures.
- Add a real bibliography with 25-40 relevant citations before submission. The current reference list is deliberately minimal and should be expanded.

Appendix L. How to write the final rebuttal-ready contribution paragraph

A concise rebuttal-ready contribution paragraph should read as follows: We do not claim that emotion categories are psychologically pure or that LLMs reproduce neural emotion maps. We propose an operational geometry of ambiguity in labelled representation spaces. The key empirical result is that relative centroid competition, not absolute distance from a centroid, predicts uncertainty. This is shown in LLMs by tight coupling between geometric margin and logits, and in fMRI-derived ROI representations by the fact that margin predicts cross-validated classifier uncertainty and collapses under feature shuffling. Prototype purity is a secondary class-level statistic that quantifies separability within the same framework.

Appendix M. Final pre-submission audit checklist

- Abstract contains only claims that appear in figures or tables.
- Every metric is defined before first use.
- Margin sign convention is consistent in text, equations, plots, captions, and code.
- Brain analysis uses LOSO rather than random row-wise splits.
- Feature shuffle and permutation controls are reported for the main brain claim.
- LLM logit-coupling results are reported with correlations, confidence intervals, and p-value conventions.

- Purity is always described as prototype purity or centroid exclusivity, never as an intrinsic property of emotion.
- Limitations section explicitly names single-brain-dataset scope and non-causal evidence.
- Supplement includes scripts or notebooks sufficient to regenerate every figure.
- Anonymisation is checked in PDF metadata, notebook paths, figure file paths, and repository URLs.

Appendix N. Anticipated reviewer questions and concise answers

Q: Is this just clustering?

A: No. Clustering results establish that the representations contain emotion structure, but the central results link boundary geometry to logits in LLMs and to cross-validated classifier uncertainty in fMRI-derived ROI patterns.

Q: Why should centroid margin be preferred to distance?

A: Distance uses only the assigned prototype. Margin compares the assigned prototype with the nearest alternative, which is exactly where ambiguity arises. Empirically, margin outperforms distance in the brain validation and aligns strongly with LLM logits.

Q: Does the brain result prove that LLMs and brains use the same code?

A: No. The claim is local and operational: a comparable margin-ambiguity relationship is observed in both systems. The manuscript explicitly allows global representational differences.

Q: Is prototype purity a psychological claim?

A: No. Prototype purity is nearest-centroid exclusivity in a representation space. It quantifies class separability and should not be interpreted as emotional purity in subjective experience.

Q: Could the result depend on Euclidean distance?

A: This should be addressed by a sensitivity analysis using cosine distance or L2-normalised embeddings. If the result is stable, the margin principle is less metric-dependent.

Q: Why use one brain dataset?

A: The NeuroEmo dataset provides a clean five-emotion design and subject-level coverage. The paper should frame it as biological validation and list broader neural replication as future work.

Appendix O. Optional future-work paragraph

Future work should test whether centroid competition generalises beyond the present emotion labels and modalities. On the language-model side, the framework can be applied to larger model families, instruction-tuned generative models, multilingual emotion datasets, and naturalistic affective corpora. On the neuroscience side, the key next step is replication in independent fMRI, EEG, or behavioural datasets with trial-level ratings. A particularly strong extension would pair subjective emotion-intensity reports with neural representations, allowing geometric margin to be evaluated against human-reported ambiguity rather than classifier-derived uncertainty alone.

A second direction is causal intervention. In language models, one can perturb embeddings along margin-related directions, remove competitor directions, or steer samples away from boundary regions and test whether logits change accordingly. Such experiments would move the paper from predictive geometry toward causal representation editing. In brain data, causal claims are harder, but stronger evidence could come from within-subject designs, trial-level mixed-effects models, and convergent validation across stimulus sets.